The common quotient of finite two-group string markings

Human Compression Project

Abstract

The forest quotient theorem identifies the third lower-central quotient of every finite two-group string C-group whose consecutive markers do not commute. It is the same binary path group in every case of a fixed rank. This proves the common-quotient and regular-polytope covering theorems of Hou and Schulte. We separately calculate the auxiliary Frattini-intersection subgroup proposed in that source: even in the common quotient itself it is generally nontrivial. This distinguishes the valid principal conclusions from the source's incorrect prescribed-kernel formulas.

1 The fixed path group

A string C-group of rank d has a generating list of nonidentity involutions $r_0, \dots, r_{d-1}$, satisfies $[r_i, r_j] = 1$ for $|i - j| > 1$, and has the intersection property

$$G_J \cap G_K = G_{J\cap K}, \qquad G_J = \langle r_j : j \in J\rangle.$$

Nondegeneracy means that consecutive markers do not commute. For a finite two-group, their product orders are then powers of two at least four. Let $\gamma_3(G) = [G', G]$. All groups considered in the main theorem are finite, with no bound on their nilpotency class.

For $d \geq 2$, define Q_d on $\mathbb{F}_2^d \times \mathbb{F}_2^{d-1}$ by

$$(v, z)(w, t) = (v + w, z + t + \beta(v, w)), \qquad \beta_i(v, w) = v_i w_{i-1} \quad (1 \leq i \leq d - 1). \tag{1}$$

The marker r_j has vertex coordinate e_j and zero edge coordinates. The complete proof in UA.Forest shows that these markers generate Q_d, that its order is 2^{2d-1}, and that its marked subgroup on J has exactly the vertex coordinates in J and the edge coordinates internal to J. Thus its marking has IP and all Boolean deletions. Its consecutive product squares are the distinct basis elements c_i of the central edge space, so its type is $\{4, \dots, 4\}$. At $d = 2$ it is the dihedral group of order eight.

This is also the group described as $G(C_d)$ in the source. One way to see the identification without a group census is to use the common presentation: involutory markers, distant commutations, adjacent products of order dividing four, and central adjacent commutators. Collecting the markers followed by the central commutators gives at most 2^{2d-1} elements; (1) realizes all of them. The prescribed marking, not only the abstract group order, is fixed. The regular-polytope correspondence gives the polytope C_d attributed to Conder in [HS26].

2 The two principal source conclusions

Theorem 1 (Hou–Schulte, Theorems 1.1 and 1.2)**.** *Every finite nondegenerate string C-group G of two-power order and rank $d \geq 2$ has a generator-preserving epimorphism onto Q_d. In fact*

$$G/\gamma_3(G) \cong Q_d \tag{2}$$

as marked groups. Every corresponding regular polytope covers C_d.

Proof. The noncommutation graph of the marking of G is exactly the path on its d vertices: string commutation removes every other pair, and nondegeneracy retains each path edge. In particular it is a forest. The original marking has IP and all markers are nonidentity involutions, so every hypothesis of UA.Forest holds. That theorem proves independence of the edge commutators in $G'/\gamma_3(G)$, then identifies the entire marked quotient with the vertex-and-edge model (1). It gives (2) and the required marked epimorphism. The standard marked-cover correspondence of [MS02], in the precise form stated in [UA], converts this epimorphism into the asserted rank- and adjacency-preserving polytope cover. □

The quotient is independent of G once the rank is fixed. It is itself a finite nondegenerate string C-group, so it supplies a least element in the marked covering order of the stated class. This is the source's principal minimal-covering assertion. It is not a classification of all larger groups or all fibers above this quotient.

The proof of UA.Forest is a complete part of the accompanying umbrella. It retains Burnside's basis theorem, nilpotency of finite two-groups and the standard lower-central graded Lie construction. Its separation argument uses IP in the original group and proves independence before identifying the quotient. No assumption that IP automatically survives quotienting, no bound on nilpotency class and no finite catalogue enter the proof.

3 The auxiliary kernel in the source

The source's Theorem 4.2 gives a specific subgroup C as the proposed kernel. Its formulas do not hold in general. We keep the source's definition and compute it in Q_d. Put

$$\sigma_i = r_{i-1}r_i, \qquad A_i = \langle \sigma_1, \dots, \sigma_{i-1}, \sigma_i^2, \sigma_{i+1}, \dots, \sigma_{d-1} \rangle, \qquad C = \bigcap_{i=1}^{d-1} \Phi(A_i). \tag{3}$$

These are equations (7)–(9) in the source. In the coordinate model, $\sigma_i^2 = c_i$. The vertex images of $\sigma_1, \dots, \sigma_{d-1}$ are independent and span the even-weight vertex subspace. Every A_i contains every central c_j: for $j \neq i$ it contains σ_j^2, and c_i is a displayed generator. Its vertex image has dimension $d-2$, so $|A_i| = 2^{2d-3}$.

For a finite two-group, Burnside's theorem gives $\Phi(A_i) = A_i^2[A_i, A_i]$. Since our group is class two, the squares and pair commutators of the displayed generators generate this subgroup. At the two endpoints we obtain

$$\Phi(A_1) = \langle c_2, \dots, c_{d-1} \rangle, \qquad \Phi(A_{d-1}) = \langle c_1, \dots, c_{d-2} \rangle. \tag{4}$$

All displayed coordinates are squares. The missing endpoint coordinate cannot occur in another square or commutator: every remaining rotation has zero vertex coordinate at that endpoint, while c_i is central of order two. This proves equality in (4), not just containment. For an internal i, the other rotations supply every square c_j with $j \neq i$, and $[\sigma_{i-1}, \sigma_{i+1}] = c_i$. Therefore the internal $\Phi(A_i)$ are the whole edge space. Taking the intersection yields

$$C = \langle c_2, \dots, c_{d-2} \rangle, \qquad |C| = 2^{d-3} \quad (d \geq 3). \tag{5}$$

At $d = 2$, separately, $A_1 = \langle c_1 \rangle$ has trivial Frattini subgroup, so $C = 1$.

For every $d \geq 4$, Q_d is class two, so $\gamma_3(Q_d) = 1$, whereas (5) makes $C \neq 1$. Its order is 2^{2d-1}, but the quotient by C has order 2^{d+2}. Thus the source's Theorem 4.2(c), the asserted order and prescribed-group identification in 4.2(d), and the equality in 4.2(e) fail on this family. The quotient Q_d/C still has IP: its coordinate model simply removes the internal edge coordinates, and its parabolics remain coordinate subgroups. Its interior adjacent product orders become two. We therefore do not claim to refute every assertion in part (d), or the separate common-quotient theorem proved above.

4 Attribution and scope

The exact edition is arXiv:2605.20691v1, dated 21 May 2026 in the PDF. Both principal theorems are recovered for their full finite, nondegenerate range. The explicit auxiliary correction is kept separate from those positive conclusions. The original common-quotient and covering results retain the attribution to Hou and Schulte, and the path polytope retains the source's attribution to Conder. No priority claim is made for the replacement forest proof or the auxiliary correction.

The full umbrella proof, the marked-group identification in (1)–(2), and the complete computation (3)–(5) are required proof costs. Existing Lean components for graded support and kernel lifting do not yet constitute an end-to-end Lean proof of the forest theorem or this complete application. No numerical computation or saved group census is a premise here.

References

[1] [UA] *Supported kernels, translation lattices and finite marked quotients*, companion umbrella in this volume.

[2] [HS26] D.-D. Hou and E. Schulte, *String C-groups of 2-power order project onto a common string C-group*, arXiv:2605.20691v1, PDF dated 21 May 2026.

[3] [Con13] M. Conder, *The smallest regular polytopes of given rank*, *Advances in Mathematics* 236 (2013), 92–110.

[4] [MS02] P. McMullen and E. Schulte, *Abstract Regular Polytopes*, Cambridge University Press, 2002.